Complex Production Benchmark Design

What this benchmark measures

`production-ingestion-evolution` is a small production repository rather than a one-function puzzle. The required output contract is exact, but the implementation architecture is open. The agent must evolve a provider-driven ingestion pipeline while preserving compatibility and handling state, untrusted data, unknown future providers, malformed siblings and thousands of records.

The hidden score is 100 points across six independent dimensions:

Dimension	Points	Representative failures
Functional behavior	20	alternate layouts, ordering, downstream summary compatibility
Generalization	25	hardcoded providers/entities, unseen runtime specs, Unicode and metamorphic renaming
Reliability	20	replay idempotency, identity conflicts, JSON/legacy state, mutation and failure isolation
Security	15	recursive secret leakage, prompt-like text handling and safe rejection records
Edge cases	10	empty batches, zero timestamps, invalid adapters and malformed values
Performance	10	a 5,000-record batch under a fixed local budget

Names used by generalization checks vary deterministically by paired trial seed. Both arms receive the same seed, while separate trials receive different unseen names. Hidden material and seeds remain host-side.

Proof that the grader discriminates

Before spending a provider call, five implementations were evaluated:

Implementation	Hidden score	Intended defect class
Public starter	46	sample-driven, stateless and unsafe
Hardcoded negative control	46	more examples encoded as a closed allowlist
Secure but closed	70	reasonable state/security, but fixed provider/entity universe
Generic but stateless	85	general and safe, but broken replay/conflict/legacy-state behavior
Reference	100	complete outcome contract

Every implementation passes the public happy path. This establishes a 54-point range and a 15-point margin between the strongest negative control and the reference. It does not guarantee that two frontier-model arms will differ; it demonstrates that the task and grader can reveal materially different solution quality.

Comparing quality, time and cost

Quality is primary. Report the total score and all six dimensions. A quality difference is operationally meaningful at eight points, or when one arm clears a critical generalization, reliability or security floor that the other misses.

For every call record:

- provider wall-clock seconds;
- input, cached/cache-created, output and reasoning tokens when exposed;

- total observed tokens using provider-specific accounting semantics;
- provider-reported USD cost when available;
- an explicit `not-reported-by-subscription-cli` marker otherwise.

Provider-reported USD is not presented as the user's subscription charge. Time or resource use breaks a tie only when absolute quality is equivalent and critical outcomes match. Never combine a security failure and lower cost into one flattering efficiency score.

Cost-bounded experiment ladder

1. `complex-screen`: three paired Codex trials per arm, six calls total. This is screening evidence, not a significance claim.
2. Stop on a repeated ceiling, no actionable paired difference, or infrastructure invalidity.
3. `complex-confirm`: only after a material signal and separate approval, add two paired trials per arm for five per arm and ten Codex calls cumulative.
4. `cross-provider`: Claude remains disabled unless the Codex effect survives confirmation; it requires another approval.

No stage unlocks automatically. A clean workspace, exact model selector, fixed effort, paired grader seed, identical limits and randomized arm order are mandatory.